BREATHTWIN (DIGITALTWIN) // SYSTEM & CODEBASE ARCHITECTURE DIAGRAM

HELD-OUT TEST ROC-AUC: 24H = 0.9399 | 48H = 0.9428

Three-Stream Data Fusion | Mechanistic Airway ODE Simulator | Stacked Dilated-TCN + XGBoost + MLP Ensemble | FastAPI & Next.js 16

STREAM 01 - STATIC EHR (60 FEATURES)

Patient Clinical Baseline

- Demographics, BMI, Smoking Pack-Years
- Spirometry: FEV1%, FEV1/FVC, PEF
- ICS Controller Regimen & Adherence
- Prior Exacerbations & ICU History

Path: aasthma/dataset/data/static/

STREAM 02 - WEARABLE TELEMETRY (20 CH)

12-Hour Sliding Sensor Window

- SpO2 (%), Resting HR, HRV (rMSSD)
- Respiratory Rate, Skin Temp, Steps
- Sleep Awakenings & SABA Inhaler Puffs
- Cough, Wheeze, Dyspnea Scores (0-3)

Path: aasthma/dataset/data/dynamic/

STREAM 03 - ENVIRONMENT (CPCB / AQI)

Ambient Air Quality & Weather

- Hourly PM2.5, PM10, AQI Index
- 70+ Indian Monitoring Stations
- Simulation Replay Clock (t=12..239h)

Path: aasthma/simulation/clock.py

STORAGE - POSTGRESQL 16 + SQLALCHEMY

Relational & Telemetry Persistence

- patients & patient_history tables
- timeseries_telemetry (indexed by hour)
- clinical_events & severity grades
- patient_daily_logs (PRO check-ins)
- user_accounts (Doctor & Patient RBAC)

Path: aasthma/database/models.py

STAGE 1A - DEEP TEMPORAL

Dilated TCN Network

- StaticMLPEncoder (60->64D)
- Conv1DDynamicEncoder (20x12h)
- Dilated Conv1D (d=1, 2, 4)
- Heads: P(24h), P(48h), Sev

model1/digital_twin_model.pt

STAGE 1B - BOOSTED TREES

XGBoost Multi-Target

- Rolling mean, std, min, max
- Temporal lag shifts (t-2..t+2)
- Static EHR fusion vector
- Heads: P(24h), P(48h), Sev

model2/gb_digital_twin_models.pkl

STAGE 2 - STACKING META-LEARNER & CONFORMAL CALIBRATION

RespiTwinMLPMetaLearner (12D -> 32D -> 16D -> 3 Heads)

- Fuses 12 base probabilities from Stage-1 TCN & XGBoost
- Computes 90% Conformal Risk Intervals + SHAP Attributions
- Held-Out Test ROC-AUC: 24h = 0.9399 | 48h = 0.9428 | F1 = 0.8435

Path: aasthma/training/ensemble.py | aasthma/inference/inference_engine.py

MECHANISTIC TWIN SIMULATOR - LATENT AIRWAY ODE

dI/dt = -k*I + a*Dose_PM(t) + b*Trigger(t) + d_sys - c*Controller(t)

- Euler-Maruyama ODE integration (dt = 1.0h) for inflammation I(t)
- Counterfactual What-If: Indoors, N95 (95% filter), HEPA, ICS Step-Up
- Re-scores counterfactual trajectories through neural stack (<65 ms)

Path: aasthma/simulation/mechanistic_simulator.py

STAGE 3 - CLINICAL COPILOT & NLP NOTE EXTRACTOR

Pydantic-Validated LLM Copilot (GPT-4o / HF Llama-3.1 / Qwen2.5)

- Generates structured CopilotReport (precautions, triggers, red flags)
- Extracts free-text clinical notes (Typhoid, Dengue) into ODE deltas
- Non-prescriptive safety guardrail: clinician retains medication control

Path: aasthma/llm/llm.py | aasthma/llm/clinical_extractor.py

BACKEND API - FASTAPI (:8000)

REST Gateway & Replay Clock

- GET /api/patients & /{id}
- POST /api/simulate/what-if
- POST /api/patients/{id}/adapt-note
- POST /api/patients/{id}/copilot-report
- POST /api/auth/login|signup|verify

Path: aasthma/server.py

DOCTOR UI - NEXT.JS 16 + THREE.JS

Clinician Triage & Twin Viewport

- /patients: Cohort Risk Triage
- /patients/[id]: 3D GLB Lung & GLSL Thermal Inflammation View
- /patients/[id]/forecast: 24h/48h Conformal Fan Chart & Copilot
- /patients/[id]/what-if: Sandbox
- /patients/[id]/adaptation: Notes
- /benchmarks & /assumptions

Path: frontend/app/ & components/

PATIENT UI - PORTAL (:3000)

Personal Asthma Companion

- Role-isolated patient login
- Plain-language 24h/48h risk card
- Local CPCB station AQI advisory
- Daily symptom & SABA puff log
- Grounded AI patient copilot chat

Path: frontend/app/patient-portal/